SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	26-10-2026	**Duration:**	60 minutes

Code of Conduct
I Jai Ganesh Rao N (192524073) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: Jai Ganesh Rao N

Annexure A – Architecture Design Assignment

Instructions to Students:
Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Dark web threat actor de-anonymization

1. Introduction:

The Dark Web Threat Actor De-anonymization and Intelligence Platform is a cybersecurity solution designed to assist authorized investigators in analysing threat intelligence and identifying potential connections between suspicious online identities. The system uses Artificial Intelligence (AI) to analyse indicators such as usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure information.

The proposed architecture integrates threat intelligence collection, data processing, AI-assisted relationship analysis, secure data storage, and visualization into a single platform. It aims to reduce manual investigation efforts, improve the organization of threat intelligence, and support evidence-based investigations while ensuring that potential connections are verified by human analysts.

2. System Requirements Analysis:

2.1 System Objectives

The main objectives of the Dark Web Threat Actor De-anonymization and Intelligence Platform are:

- To collect threat intelligence from authorized sources.
- To extract indicators such as usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure details.
- To identify potential relationships between suspicious online identities.
- To use AI-assisted analysis to detect patterns and connections.
- To visualize relationships through graphs and activity timelines.
- To generate intelligence reports for cybersecurity investigations.
- To maintain secure data storage and traceable investigation records.

2.2 Functional Requirements

Functional requirements describe the operations the system must perform.

ID	Requirement	Description
FR1	User Authentication	Allow authorized users to log in securely.
FR2	Intelligence Collection	Collect available intelligence from authorized sources.
FR3	Indicator Extraction	Extract relevant threat indicators from collected data.
FR4	Relationship Analysis	Identify potential connections between online identities.
FR5	AI-Assisted Analysis	Analyse patterns and suggest possible relationships.
FR6	Visualization	Display relationships using graphs and timelines.
FR7	Investigation Management	Allow investigators to manage investigation records.
FR8	Report Generation	Generate intelligence reports in supported formats.
FR9	Audit Logging	Record important user actions and investigation activities.

2.3 Non-Functional Requirements

Non-functional requirements describe the quality and performance expected from the system.

ID	Requirement	Description
NFR1	Scalability	Support increasing amounts of intelligence data.
NFR2	Security	Protect sensitive data through authentication and access control.
NFR3	Performance	Process data and display results efficiently.
NFR4	Reliability	Maintain accurate records and handle system failures.
NFR5	Maintainability	Allow components to be updated independently where practical.
NFR6	Availability	Provide reliable access to authorized users.
NFR7	Usability	Provide a simple interface for investigators.
NFR8	Data Integrity	Preserve the accuracy and traceability of collected evidence.

2.4 Key Quality Attributes

The architecture must consider six important quality attributes:

- Scalability: Handles growth in data and user numbers.
- Security: Protects sensitive investigation information.

- Performance: Processes intelligence and displays results efficiently.
- Reliability: Preserves investigation records and supports dependable operation.
- Maintainability: Makes the system easier to modify and extend.
- Availability: Keeps the platform accessible when needed.

These requirements will guide the selection of a suitable software architecture.

3. Selection of Suitable Software Architecture

3.1 Selected Architecture: Hybrid Architecture

The Hybrid Architecture combining Layered Architecture and Microservices is suitable for the Dark Web Threat Actor De-anonymization and Intelligence Platform. It separates the system into layers while allowing major functions, such as intelligence collection, AI-assisted analysis, and report generation, to operate as independent services.

3.2 Architecture Components

The proposed architecture consists of the following layers:

Layer	Components	Purpose
Presentation Layer	Web Dashboard, Relationship Graphs	Provides the user interface for investigators.
Application Layer	Authentication, Investigation Management, Report Generation	Manages user requests and system operations.
Intelligence Processing Layer	Data Collection, Indicator Extraction, Relationship Analysis	Processes collected threat intelligence.
AI Analysis Layer	Pattern Analysis, Confidence Scoring	Suggests potential relationships between online identities.
Data Layer	Database, Evidence Records, Audit Logs	Stores intelligence data and investigation records securely.
External Interface Layer	Authorized Intelligence Sources, External APIs	Supports permitted data collection and integration.

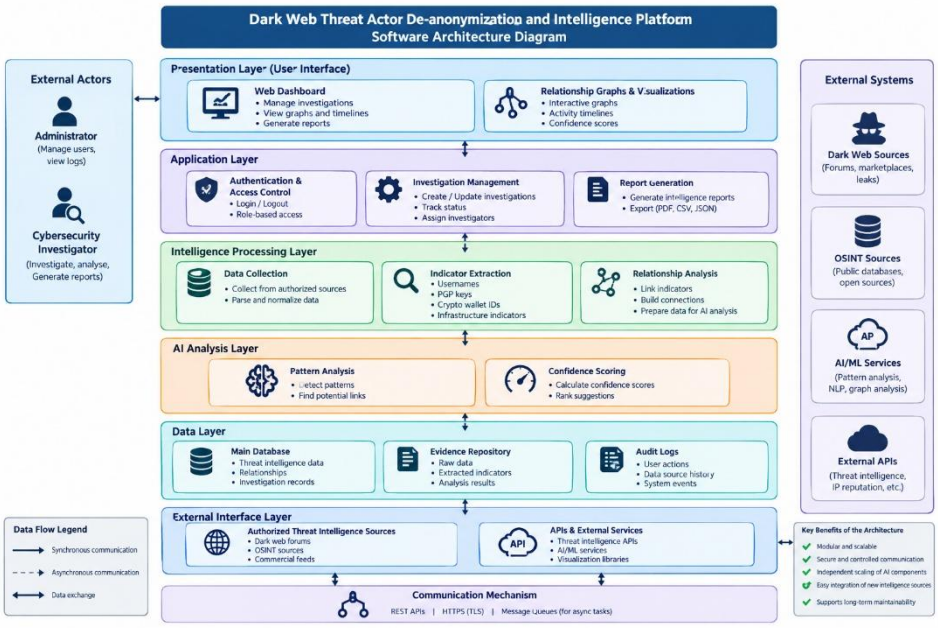
3.3 Justification for Selecting Hybrid Architecture

The hybrid approach is suitable for the following reasons:

1. Modularity: Separates system functions into manageable components.
2. Scalability: Allows resource-intensive components, such as AI analysis, to scale independently when implemented as separate services.
3. Maintainability: Individual components can be updated with limited impact on other components.
4. Security: Supports centralized authentication, access control, and controlled communication between services.
5. Performance: Allows data processing and AI analysis to operate separately from the user interface.
6. Extensibility: Makes it easier to integrate additional authorized intelligence sources and analysis techniques.

3.4 Architectural Trade-offs

Although hybrid architecture offers flexibility, it introduces additional design and deployment complexity. Communication between services may increase latency, and distributed components require careful monitoring and security controls. For an academic prototype, the components can initially be implemented as modules within a single application and separated into independent services as the system grows.



4. Explanation of Components and Their Interactions

4.1 Component Description

The proposed architecture consists of the following components:

- Presentation Layer
  Provides a web dashboard for administrators and cybersecurity investigators to view threat intelligence, relationship graphs, timelines, and analysis results.
- Application Layer
  Manages user authentication, access control, investigation activities, and report generation. It ensures that only authorized users can access the system.
- Intelligence Processing Layer
  Collects threat intelligence from authorized sources and extracts indicators such as usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure details.
- AI Analysis Layer
  Uses AI-assisted pattern analysis to identify potential relationships between threat indicators. It assigns confidence scores to help investigators evaluate the results. AI-generated connections require human verification.
- Data Layer
  Stores threat intelligence, extracted indicators, relationship records, investigation details, evidence, and audit logs securely.

- External Interface Layer
  Connects the platform to authorized threat intelligence sources, OSINT sources, and external APIs through controlled communication interfaces.

4.2 Interaction Between Components

The components communicate through defined interfaces and APIs. The Presentation Layer sends user requests to the Application Layer. The Intelligence Processing Layer collects and processes data, while the AI Analysis Layer examines patterns and suggests possible relationships. The Data Layer stores the results, and the Presentation Layer displays them to investigators.

4.3 System Workflow

- The investigator logs in securely.
- The investigator starts an investigation and selects an authorized intelligence source.
- The Data Collection component gathers permitted information.
- The Indicator Extraction component identifies relevant threat indicators.
- The AI Analysis component examines patterns and suggests possible relationships.
- The system stores findings, evidence, and confidence scores.
- The dashboard displays relationship graphs and investigation timelines.
- The investigator verifies the findings and generates a report.
- The system records important actions in the audit log.

Discussion of Quality Attributes:

5.1 Scalability

The architecture supports adding new threat intelligence sources and increasing the volume of data. AI analysis and data collection components can be scaled independently when required.

5.2 Security

The system uses authentication, role-based access control, HTTPS communication, and secure data storage. Audit logs record important activities to support accountability and investigation tracking.

5.3 Performance

The system uses efficient data processing and database queries to reduce analysis time. Asynchronous processing can handle time-consuming intelligence collection and AI analysis without unnecessarily delaying dashboard interactions.

5.4 Reliability

The system maintains investigation records and evidence to reduce the risk of data loss. Error handling and controlled recovery procedures help components manage failures.

5.5 Maintainability

The modular architecture separates the presentation, application, intelligence processing, AI analysis, and data components. Developers can update or test individual components with minimal impact on others.

5.6 Availability

The platform can use health monitoring, backups, and recovery procedures to improve availability. Critical components can be deployed redundantly as the system grows.

6. Justification of Architectural Decisions

6.1 Architectural Justification

The proposed system uses a Hybrid Architecture combining Layered Architecture and Microservices. This approach separates the system into layers while allowing important components to operate as independent services when required.

The architecture is suitable because:

- Modularity: Each layer has specific responsibilities, making the system easier to understand and develop.
- Scalability: Components such as data collection and AI analysis can be scaled independently.
- Security: Authentication, access control, and secure communication help protect sensitive investigation data.
- Maintainability: Individual components can be updated without redesigning the entire system.
- Extensibility: New intelligence sources and analysis techniques can be integrated in the future.

6.2 Architectural Trade-offs

Although the hybrid architecture offers several benefits, it also has some limitations:

- Implementation complexity: Managing multiple services requires additional configuration and coordination.
- Communication overhead: Communication between services may increase response time.
- Deployment effort: Independent services require suitable deployment and monitoring procedures.
- Security management: Every service and communication interface must be properly secured.

For the initial academic prototype, the system can be developed as a modular application. Individual components can later be separated into independent microservices as the project grows.

6.3 Future Enhancements

The architecture can be improved in the future by introducing:

1. Advanced AI models for improved threat indicator analysis.
2. Additional authorized threat intelligence sources and APIs.
3. Real-time monitoring and threat intelligence updates.
4. Enhanced interactive relationship graphs and investigation timelines.
5. Cloud deployment for improved scalability and availability.
6. Advanced audit trails and evidence integrity verification**.**

7. Conclusion

The Dark Web Threat Actor De-anonymization and Intelligence Platform uses a hybrid architecture combining Layered Architecture and Microservices to support organized and efficient cybersecurity investigations. The system integrates threat intelligence collection, indicator extraction, AI-assisted relationship analysis, secure data storage, and visualization.

The proposed architecture improves modularity, scalability, security, and maintainability while helping investigators identify potential connections between suspicious online identities. It also supports investigation reports and audit logging. However, AI-generated relationships are only investigative leads and must be verified by authorized human investigators. Overall, the architecture provides a flexible foundation for developing a reliable threat intelligence platform, with opportunities for future enhancement and expansion.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.

Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25